Table A1.1 Systematic developments in physics over centuries (*Cont*)

▪ Birth of materials science ▪ Nanotechnology, Condensed matter physics ▪ Gravitational waves, Dark energy, Dark matter, String theory	

APPENDIX A1.2

vi) Error involving the division of two quantities (using the method of differentiation)

Consider $Z = \frac{A^n}{B^m}$

Taking logarithms of both the sides,

$$\log Z = \log A^n - \log B^m$$
$$\log Z = n\log A - m\log B$$

Differentiating both sides, we get,

$$\frac{dZ}{Z} = n\frac{dA}{A} - m\frac{dB}{B}$$

In terms of fractional error, this may be written as,

$$\pm\frac{\Delta Z}{Z} = \pm n\frac{\Delta A}{A} \mp m\frac{\Delta B}{B}$$

Maximum possible relative error in z is given by

$$\frac{\Delta Z}{Z} = n\frac{\Delta A}{A} + m\frac{\Delta B}{B}$$

vii) Error involving the product of three quantities (using calculus method)

let $u = x^m y^n Z^p$

Taking logarithms on both sides, we get

$$\log u = \log x^m + \log y^n + \log Z^p$$
$$\log u = m\log x + n\log y + p\log Z$$

Differentiating both sides, we get

$$\frac{du}{u} = m\frac{dx}{x} + n\frac{dy}{y} + p\frac{dz}{z}$$

In terms of fractional error, the equation maybe rewritten as,

$$\pm\frac{\Delta u}{u} = \pm m\frac{\Delta x}{x} \pm n\frac{\Delta y}{y} \pm p\frac{\Delta z}{z}$$

APPENDIX A2.1

Parallelogram Law of Vector addition:
If two non-zero vectors $\vec{A}$ and $\vec{B}$ are represented by the two adjacent sides of a parallelogram then the resultant is given by the diagonal of the parallelogram passing through the point of intersection of the two vectors.

Consider the figure 2.19 below. Here the two vectors $\vec{A}$ and $\vec{B}$ are connected by a common tail at an angle θ. The parallelogram OACB is next constructed.

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